

TERATOGENIC EFFECT OF DOXYCYCLINE ON GROWTH RATE OF FORE LIMB LONG BONES IN ALBINO RAT

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ABSTRACT

The possibility of association of doxycycline administration during pregnancy with bone growth abnormalities has prompted us to review the relevant experimental data.

In order to study this possible teratogenic effect of doxycycline the drug was given to pregnant female albino rats in a dose of 8mg/kg body weight intraperitoneally from day 8th to 19th of gestation. The experimental animals were obtained from the animal house of Basic Medical Sciences Institute JPMC Karachi where the study was conducted. Intact embryos were isolated by killing the pregnant female Albino rats on day's 15th to 19th of gestation and pups on 1st postnatal day.

All specimens were fixed in 80% ethanol, bulk stained with alizarin red and alcian blue and cleared in 4% KOH to reveal their bony and cartilaginous elements. The right forelimbs were disarticulated from axial skeleton and observed under stereomicroscope. The total length of humerus, ulna, radius and metacarpal bones were measured and compared with those of control animals.

The retardation in length observed at the end of experimental period in humerus was 23.62%, in ulna 25%, in radius 23.87% and in metacarpals 1-5 was 11.8, 12, 12, 11.9 and 10.5% respectively.

It was concluded that Doxycycline treatment during pregnancy might retard bone growth.

INTRODUCTION

Tetracyclines particularly Doxycycline and Minocycline are broad-spectrum bacteriostatic antibiotics commonly used for different kind of infections. These are not only prescribed by doctors but also by non-qualified persons (quacks) particularly in rural areas due to their easy availability and low cost.

Tetracyclines have been proved to be very effective in different kind of infections (Goodman & Gillman 1996) but they are associated with many toxic effects particularly when given in pregnancy. Tetracyclines cross the placental barrier (Kline *et al.* 1964) and are deposited in calcified tissues resulting in

discoloration of teeth (Kutcher *et al.* 1963) and retardation in bone growth (Cohlan 1963, Joan and Kapusnik 1996). Cohlan in 1963 reported a 40% reduction in growth of fibula. Any effect on the other long bones has not been reported. Furthermore, studies on the effect of tetracyclines have been very few in rats, which is the most commonly used laboratory animal.

The principal purpose of this paper therefore was to study the effects of doxycycline (a tetracycline antibiotic) administration during gestation on the growth of fore limb long bones in laboratory albino rat. This was done by measuring the bone length of right humerus, ulna, radius and metacarpal bones in animals whose mothers

were treated with doxycycline compared with under stereomicroscope and easy measurement of their length. length of same bones in the control animals.

Table I
Comparison of Intact Bone Length (mm) between Control and treated Albino Rats at different age periods

Bone				Parental			Postnatal
		15 th day	16 th day	17 th day	18 th day	19 th day	1 st day
Humerus	C	2.22±0.02	3.46±0.01	4.34±0.02	5.22±0.01	5.88±0.02	6.56±0.04
	E	2.06±0.02	3.05±0.00	3.71±0.01	4.32±0.02	4.74±0.01	5.01±0.04
Radius	C	1.83±0.02	2.89±0.01	3.73±0.01	4.33±0.01	4.59±0.01	5.78±0.02
	E	1.70±0.01	2.56±0.01	3.20±0.00	3.61±0.01	3.72±0.01	4.40±0.02
1 st metacarpal	C	0.21±0.00	0.28±0.00	0.35±0.00	0.41±0.00	0.48±0.00	0.59±0.00
	E	0.19±0.01	0.25±0.01	0.31±0.01	0.37±0.01	0.43±0.01	0.52±0.01
2 nd metacarpal	C	0.49±0.00	0.65±0.00	0.81±0.00	0.95±0.00	1.10±0.00	1.33±0.00
	E	0.45±0.01	0.59±0.01	0.73±0.01	0.85±0.01	0.98±0.01	1.17±0.01
3 rd metacarpal	C	0.56±0.01	0.76±0.01	0.97±0.01	1.16±0.01	1.32±0.01	1.66±0.01
	E	0.52±0.01	0.70±0.01	0.88±0.01	1.05±0.01	1.19±0.01	1.46±0.01
4 th metacarpal	C	0.51±0.01	0.67±0.01	0.87±0.01	1.05±0.01	1.22±0.01	1.51±0.01
	E	0.48±0.01	0.62±0.01	0.80±0.01	0.96±0.01	1.10±0.01	1.33±0.01
5 th metacarpal	C	0.36±0.01	0.46±0.01	0.57±0.01	0.65±0.01	0.79±0.01	0.95±0.01
	E	0.34±0.01	0.43±0.01	0.52±0.01	0.59±0.01	0.71±0.01	0.85±0.01

Groups	P value	S / N.S.	Groups	P value	S / N.S.
A1 vs B1	< 0.001	H.S.	A4 vs B4	< 0.001	H.S.
A2 vs B2	< 0.001	H.S.	A5 vs B5	< 0.001	H.S.
A3 vs B3	< 0.001	H.S.	A6 vs B6	< 0.001	H.S.

Key: C = Control; E = Experimental; S = Significant; N.S. = Non-significant; H.S. = Highly Significant

The technique employed in this study involved the bulk staining of specimens with alizarin red and alcian blue (Patton & Kaufman 1995). The material was then cleared in 4% KOH. This technique enabled clear visualization of long bone primordia

MATERIALS AND METHODS

One hundred and twenty rat fetuses and pups were included in this study. To obtain these 120 off springs, 32 healthy, 10-12 weeks of age, spontaneously cycling female

albino rats were mated with fertile male rats of same age and same strain caging, one male with two female rats (Rough 1968).

The presence of a vaginal plug (a mucoid

Right fore limbs were then disarticulated from axial skeleton at sternoclavicular joint and viewed under a stereomicroscope. The total length of humerus, ulna, radius and metacarpal bones were measured and

Table II
Comparison between growth rates of long bones (mm/day) in albino rat.

Bone		Rate between 15 th day of Gestation and 1 st Postnatal day			
		15 th Day	1 st PND	IIL	GR/D
Humerus	C	2.22±0.02	6.56±0.04	4.34±0.03	0.72
	E	2.06±0.02	5.01±0.04	2.95±0.04	0.49
Ulna	C	2.06±0.02	6.60±0.03	4.54±0.03	0.75
	E	1.86±0.01	4.95±0.04	3.09±0.03	0.51
Radius	C	1.83±0.02	5.78±0.02	3.95±0.04	0.65
	E	1.70±0.01	4.40±0.02	2.70±0.04	0.45

greenish white material) the following morning was taken as evidence of mating. This was taken as day 1 of pregnancy (Chang and Kaufman 1996), normal delivery in the rat being on day 21 (Greene 1968).

Half of the pregnant female rats were given doxycycline 8mg/kg (three times therapeutic dose) intraperitoneally from 8th to 19th day of gestation. The other half received normal saline. Intact rat fetuses were isolated on days 15,16,17,18 and 19th of gestation by killing pregnant female rats from both experimental and control groups. 10 specimens were then selected randomly from each group for each day mentioned. 10 pups were isolated from each group on day 1 of their birth. The sex of the animals studied was not considered.

All pre and postnatal specimens were killed by deep ether anesthesia and fixed in 80% ethanol after removing their skin and viscera. They were then bulk stained using alizarin red and alcian blue and finally cleared in 4% KOH and stored in glycerin (Patton and Kaufman 1995).

recorded. The mean values from control and experimental animals were compared and subjected to students "t" test. The result was compared with P chart to obtain statistical significance.

RESULTS

Intact bone length:

The mean values of intact bone length as determined by measuring the entire cartilaginous models of long bones in forelimb of control and experimental animals at days 15th to 19th of gestation and on 1st postnatal day are given in table I.

This analysis revealed that the experimental values were invariably smaller in all long bones studied as compared to the control.

On 15th day of gestation the ulna of experimental group showed a decrease in length by 10.1%, humerus by 7.2% and radius by 7.1%. 1st metacarpal bone showed a decrease in length by 9.5%, 2nd metacarpal by 8.16%, 3rd metacarpal by 7.01%, 4th

metacarpal by 5.8% and 5th metacarpal by 5.4% respectively when compared with age matched control.

On 16th day of gestation the ulna of experimental group showed a decrease in length by 13.67%, humerus by 11.8% and radius by 11.4%. 1st metacarpal bone showed a decrease in length by 10.7%, 2nd metacarpal by 9.23%, 3rd metacarpal by 7.89%, 4th metacarpal by 7.4% and 5th metacarpal by 6.5% respectively when compared with age matched control.

On 17th day of gestation the ulna of experimental group showed a decrease in length by 16.0%, humerus by 14.5% and radius by 14.2%. 1st metacarpal bone showed a decrease in length by 11.4%, 2nd metacarpal by 9.87%, 3rd metacarpal by 9.27%, 4th metacarpal by 8.0% and 5th metacarpal by 8.7% respectively when compared with age matched control.

On 18th day of gestation the ulna of experimental group showed a decrease in length by 18.1%, humerus by 17.2% and radius by 16.6%. 1st metacarpal bone showed a decrease in length by 9.7%, 2nd metacarpal by 10.5%, 3rd metacarpal by 9.48%, 4th metacarpal by 8.5% and 5th metacarpal by 9.2% respectively when compared with age matched control.

On 19th day of gestation the ulna of experimental group showed a decrease in length by 20.1%, humerus by 19.3% and radius by 18.9%. 1st metacarpal bone showed a decrease in length by 10.4%, 2nd metacarpal by 10.9%, 3rd metacarpal by 9.84%, 4th metacarpal by 9.8% and 5th metacarpal by 10.1% respectively when compared with age matched control.

On 1st postnatal day the ulna of experimental group showed a decrease in length by 25.0%, humerus by 23.2% and radius by 23.8%. 1st metacarpal bone showed

a decrease in length by 11.8%, 2nd metacarpal by 12.0%, 3rd metacarpal by 12.0%, 4th metacarpal by 11.9% and 5th metacarpal by 10.5% respectively when compared with age matched control.

Growth rate of long bones:

The growth rate of long bones was calculated by measuring the difference between their length on final (1st post natal day) and initial (15th day of gestation) days of experiment divided by the number of days. The values are given in table II.

Accordingly during the study period the ulna, the humerus and the radius showed an increase in length at a rate of 0.51, 0.49 and 0.45 mm per day in the experimental animals and 0.75, 0.72 and 0.65 mm per day in the control animals showing a decrease by 32%, 31.9% and 30.7% respectively in experimental animals.

DISCUSSION

An examination of the literature regarding the teratogenic effects of tetracyclines shows contrasting results. Some workers describe maldevelopment in some bones (e.g. fibula), others have not observed such abnormality after similar treatment. In general the higher the dose the more is the teratogenic effect (Gennazani *et al.* 1965).

In this study we have investigated the effect of doxycycline administration during gestation on growth of forelimb long bones in albino rat.

A detailed examination revealed that doxycycline retarded the growth of these long bones. The effect was however varied in different bones. Ulna showed retardation by 25% in its length, humerus by 23.62% and radius by 23.87%. This means that the growth retarding effect was relatively more marked in distal long bones of the limb than the proximal bones. This may be attributed to the

fact that during gestation the distal long bones grow faster than the proximal bones (Arey 1974) resulting into more retardation in them.

This growth retarding effect of tetracyclines may be attributed to their effect on protein synthesis. They inhibit protein synthesis in microorganisms by binding with 30S sub unit of their ribosome and thus preventing the assembly by aminoacyl tRNA to its acceptor site at ribosome-mRNA complex. They may have similar effect on mammalian cell in higher doses (Joan & Kapusnik 1996). Protein synthesis is an essential aspect of laying down ground substance in bones as well as in cartilage and if inhibited may adversely affect formation of bone.

Doxycycline has also been reported to disrupt chondrocyte differentiation and inhibit calcified cartilage degradation (Cole et al. 1994). These steps are essential features of long bone growth in length and if inhibited may also adversely affect the growth in long bones.

Doxycycline also inhibits the activity of metalloproteinases (collagenases & gelatinases) (Sorsa 1993), which are responsible for the degradation of proteoglycan and collagen of the ground substance. This may be a contributing factor in its bone growth retarding effect.

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